ML for DIABETES DETECTION on the PIMA DATASET

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1 INTRODUCTION

Diabetes is a group of chronic and non-communicable metabolic diseases that affect how the body uses blood sugar. It is characterized by either a high blood sugar level due to insufficient insulin production in the body or ineffective use of insulin. In this project, some machine learning (ML) techniques are used to analyze PIDD (Pima Indian Diabetes Dataset) for the sake of identifying the risk factors of diabetes mellitus.

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2 DATASET DESCRIPTION and MORE

We have used the Pima Indian Diabetes Dataset (PIDD). The data is collected from 768 female patients of Pima Indian heritage and they are 21 years old or older. The dataset is arranged in 9 features which are summarized in Table 1:

Table 1: Features of PIDD and their meaning.

Feature	Meaning
Pregnancies	Number of pregnancies
Glucose	Plasma glucose concentration
BloodPressure	Diastolic BP (mm Hg)
SkinThickness	Triceps skin-fold thickness
Insulin	2-hour serum insulin
BMI	Body mass index
DiabetesPedigreeFunction	Genetic pre-disposition
Age	Age in years
Outcome	Diabetes status (1: diabetic, 0: non-diabetic)

Moreover, we found out that 35% of the patients are diabetic while the remaining are non-diabetic, thereby signaling the existence of a moderate imbalance in the data. The specific patterns of the underlying dataset are presented in Table 2.

Table 2: PIDD specific patterns.

Feature	Diabetics	Non-Diabetics	Insight
Glucose	Much higher	Lower	Strongest early signal
BMI	Higher	Lower	Obesity risk
Age	Higher	Lower	Risk increases with age
Pregnancies	Higher	Lower	Gestational risk
Pedigree	Higher	Lower	Genetic component

3 DATA INSPECTION

At a first glance, it seems that there are no NaNs which could be misleading. There are also a lot of zeros (see Table 3) in the data which makes us a bit confused, as several medical features cannot be zero physiologically. Since we are dealing with PIDD, we need to watch out for the invalid zeros in Glucose, BloodPressure, SkinThickness, Insulin and BMI. As suspected, we found a very important information which we summarize in the table below. This table tells us that, among other things, nearly 30% of SkinThickness and 49% of Insulin go missing. Properly treating these incidents will dramatically improve model reliability where as ignoring it will definitely lead to false predictions. Data imputation is needed here to properly deal with the missing values. Since median is robust to skewed medical variables and a best practice in this domain, all the missing values are replaced by the median of their respected features.

Table 3: Invalid zeros in PIDD.

Feature	Number of invalid zeros
Glucose	5
BloodPressure	35
SkinThickness	227
Insulin	374
BMI	11

4 INTERPRETING RISK FACTORS with LR and RF

The `LR` model provided the risk factors in descending order as depicted in Figure 1:

```
Glucose                    1.183444
BMI                        0.709709
Pregnancies                0.373019
DiabetesPedigreeFunction   0.287745
Age                        0.186393
Insulin                   -0.044651
BloodPressure             -0.014548
SkinThickness              0.013877
dtype: float64
```

Figure 1: Risk factors PIDD.

The results depicted in Figure 1 tell the following story:

- Glucose $-->$ strongest predictor
- BMI + Age $-->$ compounding risk
- DiabetesPedigreeFunction $-->$ genetic susceptibility
- Coefficients explain direction and strength of risk

Next, we change the model to the non-linear `RandomForestClassifier()` just to improve performance, see Figure 2:

```
[[85 15]
 [24 30]]
              precision    recall  f1-score   support

           0       0.78      0.85      0.81       100
           1       0.67      0.56      0.61        54

    accuracy                           0.75       154
   macro avg       0.72      0.70      0.71       154
weighted avg       0.74      0.75      0.74       154

ROC-AUC: 0.8160185185185184
```

Figure 2: Calculations with `RandomForestClassifier()` for PIDD.

The results in Figure 2 tell us that RF:

- Captures nonlinear interactions (e.g., Age × BMI)
- Often improves recall slightly
- Less interpretable but stronger predictive power

We have the following key early detection conclusions:

- Glucose is the single most powerful early indicator
- BMI and Age amplify risk together
- Genetic predisposition is non-negligible
- Recall-optimized models reduce missed diabetes cases significantly

5 FEATURE DISTRIBUTION

For PIDD, the most informative "relative plots" are feature distributions split by Outcome (0 vs 1). These let us visually compare diabetic vs non-diabetic patients feature by feature, which is exactly what we want for early detection insight. Since KDE is ideal for medical risk comparison, we have plotted KDE distributions for each feature, separated by Outcome as shown in Figure 3.

Figure 3 depicts the relative density distribution of the features by outcome. The interpretation is done by looking at the rightward shifts and separation of the curves; orange (diabetic) and blue (non-diabetic):

- **Glucose:** Most important feature, diabetic curve shifts strongly right and minimal overlap at high glucose. These imply glucose is the best early detection marker.
- **BMI:** Diabetics concentrated at higher BMI, clear obesity-related risk. This means, BMI is an important modifiable risk factor.

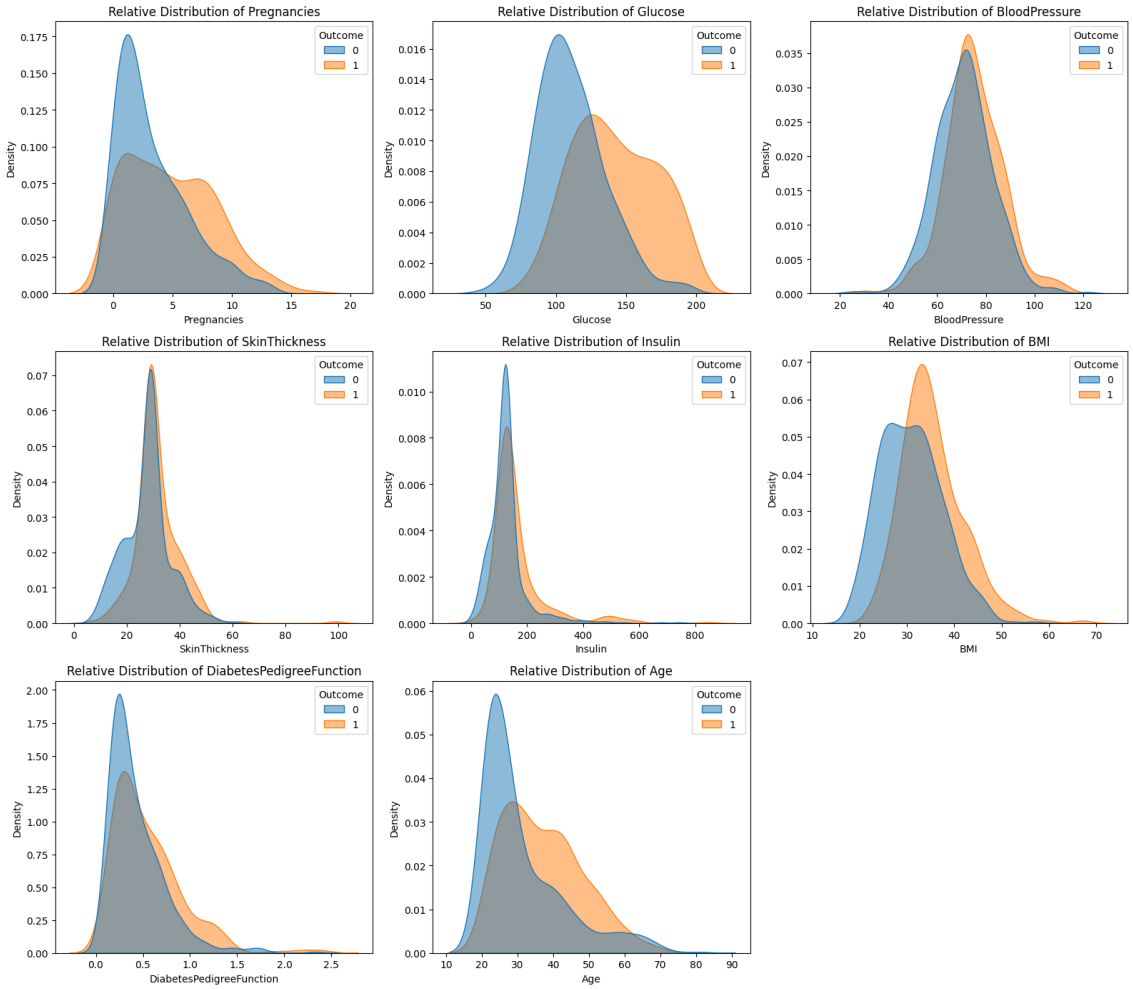

Figure 3: Relative density distribution plot.

- **Age:** Diabetes probability increases with age, older tail dominated by `outcome` = 1. This implies, Age is a progressive metabolic risk.
- **DiabetesPedigreeFunction:** Long right tail for diabetics, indicates genetic susceptibility. This means Pedigree is a non-obvious but important signal.
- **Insulin and SkinThickness:** Heavier overlap, still informative in non-linear models (XGBoost). They both are less predictive alone.
- **BloodPressure:** Significant overlap, weak standalone predictor. It is a secondary risk factor.
- **Pregnancies:** Higher pregnancies → higher diabetes risk, reflects gestational diabetes history. It is useful for population-specific insight.

6 XGBOOST + SHAP

We now change the model to XGBoost for a multitude of reasons including its ability to capture nonlinear effects, handle features automatically, produce strong recall score, and integrate perfectly with SHAP. Results obtained using the `XGBoost` model are depicted in Figure 4:

```
[[78 22]
 [14 40]]
              precision    recall  f1-score   support

           0       0.85      0.78      0.81       100
           1       0.65      0.74      0.69        54

    accuracy                           0.77       154
   macro avg       0.75      0.76      0.75       154
weighted avg       0.78      0.77      0.77       154

ROC-AUC: 0.830925925925926
```

Figure 4: Calculations with `XGBoost` classifier for PIDD.

Figure 4 tells us that `XGBoost` very strong for early detection because:

- Recall for the diabetes class has increased — good sign
- ROC-AUC has increased — good sign
- Fewer false negatives than LR and RF

Next, we combine `XGBoost` with `SHAP` as the merger is claimed to be the **gold** standard for explainable ML on tabular data.

7 GLOBAL FEATURE IMPORTANCE (SHAP)

Note that the train-test split was done as before and we used the training data to do summary plot of the SHAP values, see Figure 5.

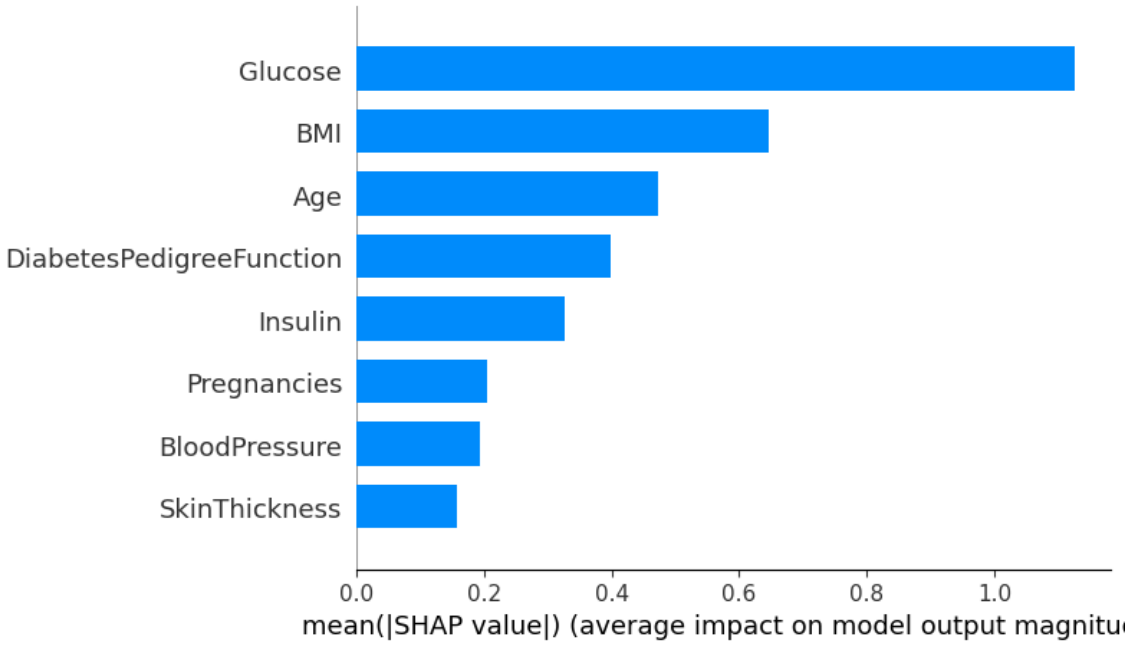

Figure 5: Summary of SHAP values obtained from `XGBoostClassifier()` + SHAP for PIDD.

The results depicted in Figure 5 tell the following ranking for the risk factors:

1. Glucose (dominant early biomarker)
2. BMI
3. Age
4. DiabetesPedigreeFunction
5. Insulin

This confirms medical plausibility, which is critical in healthcare ML.

8 DIRECTION of RISK ('BEESWARM' PLOT)

A nice looking 'bee swarm' plot is depicted in Figure 6 which is interpreted as:

- High glucose $-->$ strong push toward diabetes
- High BMI + older age $-->$ nonlinear risk amplification
- Low values $-->$ protective effect

This shows how risk changes, not just which features matter.

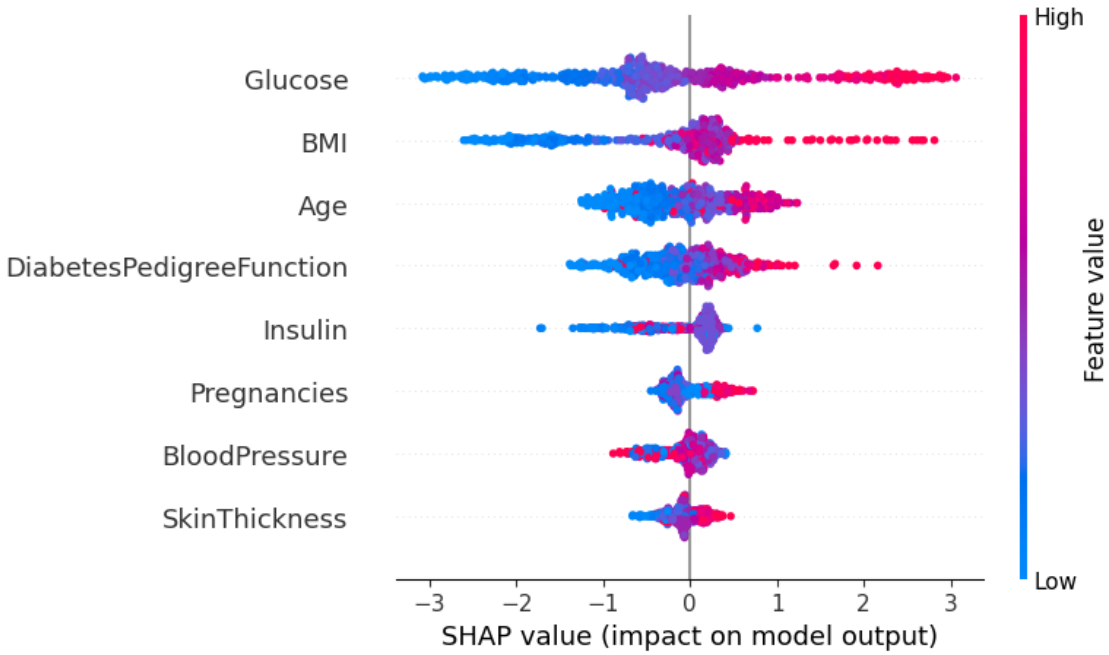

Figure 6: 'Beeswarm' plot of the summarized SHAP values using `XGBoostClassifier()` for PIDD.

9 RISK-THRESHOLD VISUALIZATIONS

Now we visualize how the risk score translates into decisions. We will use the `XGBoost` risk scores.

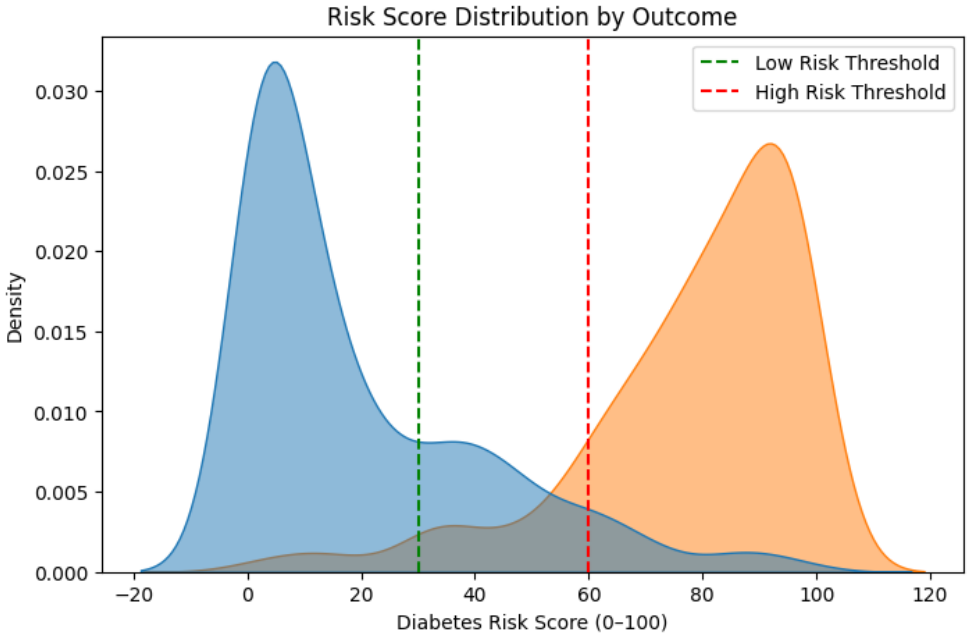

Figure 7: Risk-threshold visualizations for screening decisions.

The results in Figure 7 are interpreted as:

- Diabetics dominate right side
- Non-diabetics dominate left side
- Overlap region implies diagnostic uncertainty
- Thresholds convert probabilities into actionable rules

10 XGBOOST-BASED DIABETES RISK SCORE

The diabetic risk scores have been calculated for each individual and depicted in Figure 8 after arranging them in categories.

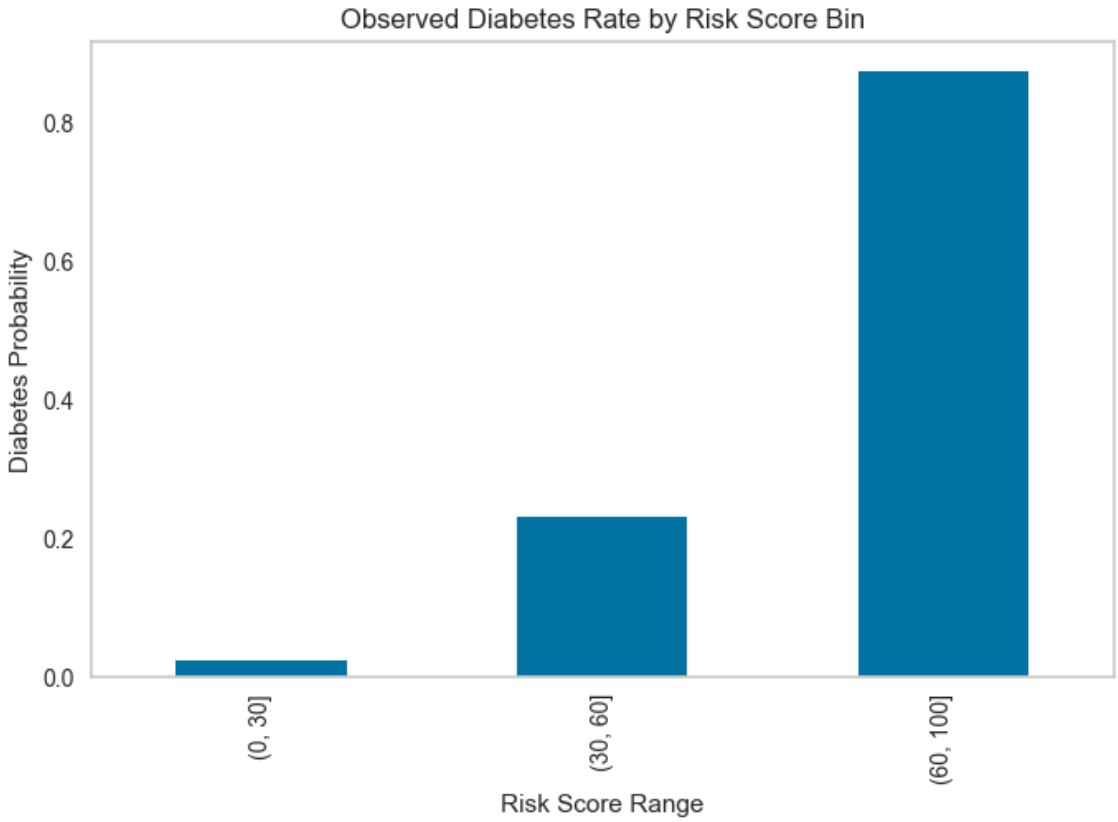

Figure 8: Risk scores vs probabilities of diabetes for PIDD.

Figure 8 tells:

- Diabetes rate increases monotonically
- High-risk bins have very high prevalence
- Confirms good calibration

- Ideal for population screening

The categorization has been done as per the risk score of the patients and summarized in Table 4.

Table 4: Summary of risk score per category.

XGB score range	Risk category	Meaning	Number of patients
< 30	Low risk	Unlikely diabetic	379
$[30, 60)$	Moderate risk	Monitor closely	265
$>= 60$	High risk	High likelihood – recommend screening	124

The risk categorization has also been validated and found out the following:

- **High-risk group** → highest diabetes prevalence
- **Low-risk group** → very low prevalence
- Confirms suitability for **screening programs**

11 INSIGHT: SCREENING POLICY and INTERVENTION

We have found the following:

1. XGBoost is found to be the winner (compared to LR and RF) because of the following reasons:

- it provides the best recall score — very useful for early detection
- it makes very good use of the non-linearity effects, i.e., combined effects of the features
- it facilitates high quality SHAP explanations
- its results are clinically trust worthy

2. Glucose is the dominant early biomarker, it pushes strongly towards diabetes
3. Combined, non-linear risk amplification is revealed by older Age + high BMI
4. The recommended screening policy is presented in Table 5.

Table 5: Summary of risk score per category.

Risk score range	Action
< 30	No action
$[30, 60]$	Monitor/lifestyle intervention
> 60	Immediate clinical testing

This policy is:

- Data-driven
- Clinically interpretable
- Early-detection optimized